

# Frequency Sweep Analysis of Thermoplastics with the RH-870

## Overview

Oscillatory frequency sweeps are a foundational rheological test for characterizing the viscoelastic properties of thermoplastic polymers. By subjecting a material to a sinusoidal shear deformation over a range of frequencies, rheologists can determine the balance between elastic storage and viscous loss behavior without altering the underlying microstructure.

This report examines the utilization of the RH-870 high-performance rheometer for mapping the complete viscoelastic spectrum of thermoplastic materials. Leveraging the advanced transducer technology, thermal stability, and extended-bandwidth capabilities of the RH-870, this analytical approach captures both rapid molecular relaxations and long-range chain dynamics in a single, continuous test sequence.

## Application

Understanding the viscoelastic spectrum of thermoplastics is critical for optimizing both processing methods and end-use performance. In manufacturing operations such as injection molding, extrusion, and blow molding, polymers experience varying deformation rates. High-frequency regimes correlate with rapid processing conditions, molecular orientation, and impact resistance, while low-frequency regimes reveal molecular weight, molecular weight distribution, terminal relaxation times, and zero-shear viscosity.

Traditional rheometers often struggle to capture the full spectrum due to torque limitations at high frequencies and compliance issues at low frequencies. The RH-870 overcomes these limitations, enabling precise measurement of the storage modulus ( $G'$ ), loss modulus ( $G''$ ), and complex viscosity ( $\eta^*$ ) across an exceptionally wide frequency window. This capability allows materials scientists to identify critical structural transitions—such as the crossover frequency, where elastic dominance supersedes viscous flow—thereby informing material selection, additive blending, and quality control protocols for commodity and engineering thermoplastics alike.

## Method

Measurements are performed using the RH-870 equipped with parallel-plate or cone-and-plate geometry, depending on the specific viscosity profile and filler loading of the thermoplastic sample. Prior to testing, polymer pellets or granules are compression-molded into uniform discs to ensure optimal contact and eliminate trapped air.

The sample is loaded into the preheated test fixture of the RH-870, where a precise normal force control mechanism ensures consistent thermal contact and squish parameters. The temperature is then stabilized above the glass transition temperature ( $T_g$ ) for amorphous polymers or above the melting point ( $T_m$ ) for semi-crystalline thermoplastics. An initial strain sweep is conducted at a fixed frequency to determine the linear viscoelastic region (LVR), ensuring that all subsequent frequency sweeps are performed well within the non-destructive deformation limit.

The extended-bandwidth frequency sweep is executed by logarithmically stepping the angular frequency ( $\omega$ ) from high to low values (typically spanning  $10^{-2}$  to  $10^2$  rad/s or wider, depending on instrument configuration and sample thermal degradation limits). Throughout the sweep, the RH-870 continuously records stress, strain, and phase angle. Temperature control is maintained via a convection oven module, ensuring isothermal conditions throughout the duration of the multi-decade sweep.

## Results

The integration of extended-bandwidth capabilities in the RH-870 yields high-fidelity rheological profiles that reveal distinct structural regimes in thermoplastic melts. At high frequencies, the storage modulus reaches a plateau corresponding to short-range segmental motions and the glassy-to-rubbery transition dynamics. Conversely, the low-frequency terminal zone accurately exposes the reptation dynamics of polymer chains, allowing for the precise extraction of zero-shear viscosity and average molecular weight characteristics.

The smooth transition between the terminal, plateau, and transition zones highlights the low inertia and high torque sensitivity of the RH-870 drive system. Furthermore, the absence of high-frequency artifacts—such as machine compliance resonance—ensures that the measured crossover modulus and frequency accurately reflect the molecular architecture of the thermoplastic. These comprehensive spectral maps provide engineers with the robust data sets required for time-temperature superposition (TTS) modeling, master curve generation, and advanced rheological structure-property correlation.